Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
A		2627068

A Physically Grounded Continuous-Time Differential Framework for Smartphone Battery Dynamics and Hardware-Level Power Decomposition

Summary

Smartphone battery endurance is a primary constraint on modern mobile computing, yet user experience varies significantly due to the complex coupling of hardware power, user behavior, and electrochemical dynamics. To predict the **State of Charge** (SOC) evolution and identify drivers of depletion, we develop a **Continuous-Time Differential Equation Model** based on physical principles rather than statistical regression.

Firstly, we establish an **Electrochemical Dynamics Sub-model** utilizing the **Thevenin/Shepherd Equivalent Circuit** to describe the battery's internal state, incorporating Open Circuit Voltage and polarization effects. To capture realistic loads, we decompose total power into six independent subsystems: Screen (based on OLED physics), CPU, Network, GPS, Background, and Base consumption. We integrate these into a master differential equation, $\frac{dSOC}{dt}$, and introduce coupled factors for battery aging and ambient temperature.

For Task 1 and Task 2, we implement the **Explicit Euler Method** to numerically solve the model, calibrating parameters based on the **Xiaomi Redmi Note 10 Pro** hardware specifications. We validate the model against **DXOMARK** datasets, where simulation results exhibit a high goodness-of-fit with the "staircase" discharge morphology of real-world mixed-use cycles. The model successfully predicts "Time-to-Empty" across varying scenarios, with relative error remaining low for the majority of the discharge cycle.

For Task 3, we perform a **Quantitative Power Decomposition** to identify "energy fingerprints". Results reveal that power consumption follows a **Pareto distribution**. The screen is the dominant consumer in standard tasks (57.5% in web browsing). We identify **High-Performance Gaming** as the primary driver of rapid drain (6.69 W), caused by the synergistic demand of GPU load and 120Hz refresh rates. Contrary to common belief, GPS hardware consumes minimal power (3.8%), with navigation drain primarily driven by associated network and screen activity.

For Task 4 and Sensitivity Analysis, we conduct a **One-at-a-Time (OAT) Perturbation Analysis**. We find that **Battery Aging** (capacity fade) is a far more critical determinant of endurance than ambient temperature, with aged batteries showing accelerated voltage sag. Based on these insights, we propose user strategies such as "Dark Mode" to leverage OLED physics and system-level optimizations like **Adaptive Refresh Rates** (LTPO). Finally, we discuss expanding the framework to Electric Vehicles (EVs) and IoT drones as a "Digital Twin" for predictive management.

Keywords: Continuous-Time Model, Differential Equations, Thevenin Equivalent Circuit, Power Decomposition, Sensitivity Analysis

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1 Introduction

1.1 Problem Background

Smartphones have become an essential component of modern life, integrating communication, navigation, entertainment, and mobile computing into a single portable device. As the functionality of smartphones continues to expand, battery endurance has emerged as a primary constraint on usability and user satisfaction. Despite advances in battery technology and energy-efficient hardware design, users frequently experience large variations in battery life under seemingly similar usage conditions.

The energy consumption of a smartphone is inherently complex. Power is drawn continuously by multiple subsystems, including the display, processor, wireless communication modules, location services, and background system tasks. These components interact dynamically over time, and their power demands depend not only on explicit user actions but also on system-level processes that operate in the background. Moreover, environmental factors such as ambient temperature, as well as battery-specific characteristics such as aging, influence both instantaneous power consumption and the effective energy capacity of the battery.

Because of these coupled and time-varying influences, predicting battery behavior using static averages or empirical heuristics is often insufficient. A meaningful analysis of smartphone battery performance therefore requires a modeling framework that captures continuous-time dynamics, incorporates physically interpretable power components, and allows systematic evaluation of different usage scenarios. Such a framework is particularly valuable for understanding why battery depletion rates differ across situations and for identifying the dominant factors that limit battery life.

Figure 1.1-1: mobile phone battery(https://www.fetnet.net/content/cbu/tw/lifecircle/tech/2023/03/charger.html)

1.2 Restatement of the Problems

The objective of this study is to develop a mathematical model that describes the discharge behavior of a smartphone battery under realistic operating conditions. The model is required to represent the battery's State of Charge (SOC) as a continuous function of time, based on the assumption that the smartphone is powered by a lithium-ion battery.

Specifically, the model should express the evolution of SOC through a continuous-time equation driven by the total power consumption of the device. The total power

demand is to be decomposed into contributions from major subsystems, including the screen, central processing unit (CPU), wireless communication modules, GPS, and background tasks, with each component modeled according to its operational characteristics. Environmental effects, such as temperature, and battery-related factors, such as effective capacity variations, should be incorporated in a physically reasonable manner.

Using the proposed model, the remaining time to battery depletion is to be estimated for different initial charge levels and usage scenarios. The model should also support comparison between predicted outcomes and observed or expected battery behavior, enabling assessment of model performance and identification of its strengths and limitations. Furthermore, sensitivity analysis is required to examine how changes in assumptions, parameters, and usage patterns affect the predicted battery life.

Finally, the modeling results are expected to yield practical insights. These include identifying which user behaviors or system conditions have the greatest impact on battery endurance, as well as providing recommendations for users or system designers to improve energy efficiency. The overall goal is to construct a transparent, extensible, and physically grounded modeling framework for analyzing smartphone battery consumption.

1.3 Our Work

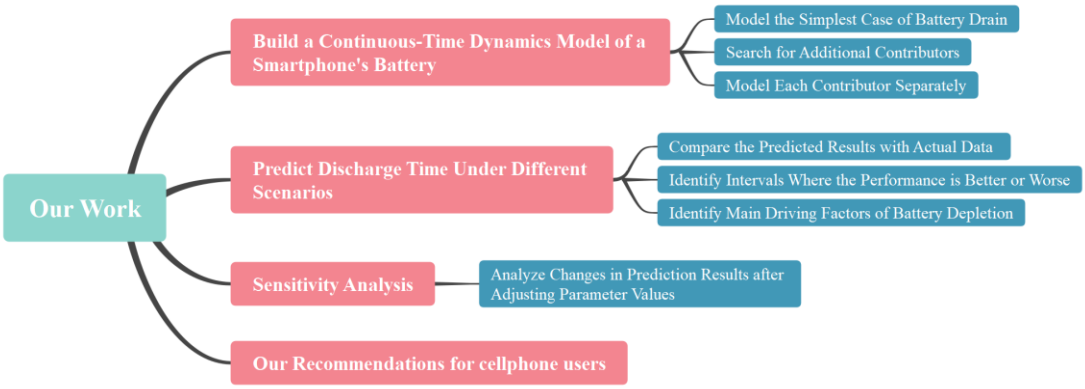

Figure 1.3-1:our work

2 Model Preparations

2.1 Assumptions and Justifications

Assumption 1: The battery is modeled as an equivalent single lithium-ion cell.
Justification: Smartphone batteries are typically composed of cells with similar structure and materials, leading to consistent macroscopic behavior.

Assumption 2: The terminal voltage of the battery is determined by the open-circuit voltage, polarization voltage, and internal resistance, with related parameters assumed constant over short time scales.
Justification: This assumption captures load-dependent voltage behavior while avoiding complex parameter dynamics, making it suitable for short-term energy

consumption modeling.

Assumption 3: Battery aging remains constant during a single discharge cycle or short-term usage.
Justification: Battery aging evolves over time scales much longer than a single discharge cycle and is primarily driven by cumulative usage and calendar time. Therefore, aging can be treated as a quasi-static parameter within short-term simulations.

Assumption 4: Battery temperature is spatially uniform and represented by a single lumped variable.
Justification: Due to the small physical size of smartphone batteries and relatively uniform heat dissipation during normal operation, internal temperature gradients are limited.

Assumption 5: Power demand is determined by usage scenarios and treated as a known external input.
Justification: Power consumption under different usage scenarios (e.g.,web browsing, video playback, gaming, and navigation) exhibits consistent statistical patterns.

2.2 Notations

Table 2.2-1:notations Description

Symbol	Meaning
SOC	State of charge
$Q_{\max}$	Rated capacity of battery
$P(t)$	Power at time t
$V(t)$	Voltage at time t
$I(t)$	Current at time t

3 Establishment of Models

3.1 Overview of the SOC Dynamic Evolution

To accurately predict the battery life, we establish a continuous-time differential equation model for the State of Charge (SOC). Instead of using a simple "black box" regression, we construct a white-box model based on the electrochemical properties of the battery and the physical power consumption of the hardware.

3.2 The Electrochemical Dynamics Sub-model

We adopt the Thevenin/Shepherd Equivalent Circuit Model to describe the battery's

internal state. The relationship between the terminal voltage $V(t)$ and the discharge current $I(t)$ is given by[7]:

$$V(t) = U_{oc}(SOC) - I(t)R_{\text{int}} - U_p(t)$$

Where: $U_{oc}(SOC)$ denotes the Open Circuit Voltage, which is a non-linear function of the current SOC, R_{int} represents the internal ohmic resistance of the battery, $U_p(t)$ that satisifies $\frac{dU_p(t)}{dt} = -0.02U_p(t) + RI(t)$ is the Polarization Voltage, representing the transient voltage drop caused by electrochemical polarization.

The rate of change of SOC is directly proportional to the discharge current[3]:

$$\frac{dSOC(t)}{dt} = -\frac{I(t)}{Q_{\max}}$$

Where $Q_{\max}$ is the rated capacity of the battery. Since the discharge current $I(t)$ is determined by the instantaneous power consumption divided by the voltage, i.e., $I(t) = \frac{P(t)}{V(t)}$, the master equation governing the system is:

$$\frac{dSOC}{dt} = -\frac{P(t)}{V(t) \cdot Q_{\max}}$$

We now introduce the impact of temperature. First, list the battery temperature equation: $C_{th}\frac{dT}{dt} = P_{loss}(t) - h(T - T_{amb})$, where $P_{loss}(t) = \eta P(t)$, C_{th} is the heat capacity, h is the heat dissipation coefficient.

Since high or low temperatures both affect battery capacity, we then examine the impact of temperature on the total battery capacity[4]: $Q_{max}(T) = Q_{ref}(1 - \beta\,|\,T - T_{ref}\,|)$.

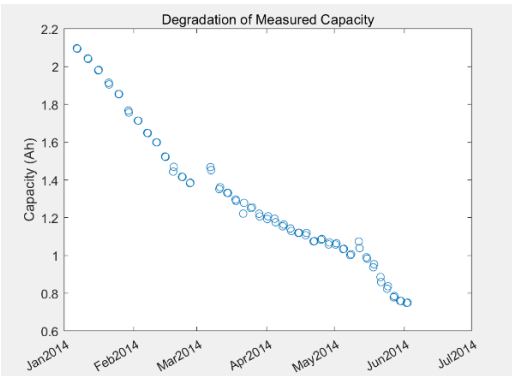

Figure 3.2-1

Also we need to consider the impact of battery aging. We have retrieved an image showing the natural aging of battery capacity over time and observed that the degree of aging appears to be approximately linear as time progresses . So we can define the aging coefficient $\alpha_{age} \in [0,1]$ to describe the battery aging[5] , then $Q_{max} = Q_{ref}(1-\alpha_{age})$.

Therefore, considering all the above factors, our main equation[6] can be modified as follows:

$$\frac{dSOC}{dt} = -\frac{P(t)}{V(t)\cdot Q_{ref}(1-\beta\,|\,T-T_{ref}\,|)\cdot(1-\alpha_{age})}$$

3.3 Detailed Power Consumption Sub-models

To solve the master equation, we decompose the power load into six distinct components[1] $P(t) = P_{base} + P_{OLED} + P_{CPU} + P_{network} + P_{GPS} + P_{backgroud}$, based on the hardware architecture described in our preliminary analysis.

(1) Basic Power Consumption (P_{base})

This represents the static power required to maintain the device in a standby state, including motherboard leakage and essential sensor operations.

(2) Screen Power Consumption (P_{OLED})

The screen is a major energy consumer. Its power depends on whether it is active and the pixel-level energy cost. We model this as:

$$P_{OLED}(t) = S_{on}(t)\cdot(P_{pixel}(t) + P_{circuit})$$

Where: $S_{on}(t) \in \{0,1\}$ is a binary variable indicating whether the screen is lit (1 for on, 0 for off), P_{pixel} denotes the power consumption related to pixel display content, $P_{circuit}$ is the fixed driving circuit power.

To further characterize the pixel-level power consumption, we adopt an empirical content-dependent model that reflects the operating principles of OLED displays. Unlike LCD screens, OLED pixels are self-emissive, and their power consumption depends primarily on pixel luminance, displayed color content, and refresh rate. Based on experimental measurements reported in prior display power studies, the pixel-related power consumption is modeled as

$$P_{pixel}(t) = 0.5\cdot L(t)^{2.2}\cdot A_{white}(t)\cdot(1+0.0083(f_{refresh}(t)-60))$$

Where $L(t)$ denotes the normalized screen luminance level, $A_{white}(t)$ represents the proportion of bright (white) pixels on the screen, and $f_{refresh}(t)$ is the display refresh rate in hertz. The nonlinear dependence on luminance reflects the gamma-

corrected relationship between perceived brightness and electrical power in OLED pixels, while the refresh-rate term captures the approximately linear increase in driving power as the refresh frequency rises above the baseline value of 60 Hz.

(3) CPU Power Consumption ($P_{CPU}(t)$)

The processor power is composed of static and dynamic parts. The dynamic power is driven by the frequency and utilization rate:

$$P_{CPU}(t) = (P_{static} + P_{dynamic}(t)) \cdot (1 + \alpha_{CPU}(T(t) - 25))$$

Where $P_{dynamic}(t) = \sum_{i=1}^{N} (P_{core,i} \cdot U_i(t)^{1.7} \cdot \frac{f_i(t)}{f_{max}})$ is the dynamic power consumption function, dependent on the CPU utilization rate and frequency.The CPU power model is derived from the classical CMOS dynamic power equation $P = CV^2 f$, extended to incorporate utilization-dependent switching activity, DVFS-induced voltage–frequency coupling, and multi-core architecture. Superlinear dependence on utilization is introduced to capture architectural inefficiencies under high load, while temperature effects are modeled using a first-order linear approximation of leakage behavior.

(4) Wireless Communication Power ($P_{network}(t)$)

The device selects one communication technology (WiFi, 4G, or 5G) at any given time. The power consumption is modeled as $P_{network}(t) = S_{tech}(t) \cdot P_{tech}(t)$

Where $S_{tech}(t)$ is a binary indicator,only when a certain mode is selected $S_{tech}(t) = 1$, $P_{tech}(t)$ means the network activity power at the current moment.In more detail,

$P_{wifi}(t) = 0.05 + 0.15 \cdot A_{network}(t)$, $P_{4G}(t) = 0.12 + 0.35 \cdot A_{network}(t)$,

$P_{5G}(t) = 0.18 + 0.45 \cdot A_{network}(t)$, where $A_{network}(t) \in [0,1]$ denotes the normalized network activity intensity at the current moment.

(5) GPS Power Consumption ($P_{GPS}(t)$)

GPS power is modeled based on the frequency of location requests rather than continuous operation.

$$P_{GPS}(t) = N_{GPS}(t) \cdot T_{session} \cdot P_{on}$$

Where: $N_{GPS}(t)$ denotes the number of GPS activations per minute, $T_{session}$ represents the duration of usage for each corresponding activation, P_{on} is the active power of the GPS hardware.

(6) Background Task Power ($P_{backgroud}(t)$)

Background processes consume energy even when the user is not interacting with the device.

$$P_{backgroud}(t) = P_{backgroud} \cdot R_{backgroud}(t)$$

Where: $P_{backgroud}$ represents the overall activity level of background tasks at the current moment, $R_{backgroud}(t) \in [1,10]$ is a scaling coefficient converting activity level to power units.

3.4 Final Continuous-Time Model

Substituting the six components back into the SOC rate equation, we obtain the final explicit differential equation for the smartphone battery:

$$\frac{dSOC(t)}{dt} = -\frac{P_{base} + P_{OLED} + P_{CPU} + P_{network} + P_{GPS} + P_{backgroud}}{Q_{ref}(1-\beta|T-T_{ref}|)\cdot(1-\alpha_{age})\cdot V(t)}$$

This equation allows us to simulate the continuous trajectory of the battery level under varying usage behaviors defined by the variables S_{on}, N_{GPS}, and $R_{backgroud}$.

4 Problem Solutions

4.1 Task1: SOC Dynamics Simulation

Based on the model development process from earlier, we can derive the following system of main equations:(more details on parameters and expressions in the model development section)

$$\begin{cases} \dfrac{dSOC}{dt} = -\dfrac{P(t)}{V(t)\cdot Q_{ref}(1-\beta|T-T_{ref}|)\cdot(1-\alpha_{age})} \\ V(t) = U_{oc}(SOC) - I(t)R_{\text{int}} - U_p(t) \\ C_{th}\dfrac{dT}{dt} = P_{loss}(t) - h(T-T_{amb}) \end{cases}$$

First , we combine the Thevenin/Shepherd-type equivalent electrochemical model with the current-voltage-power relationship to establish a system of equations. The electrochemical model provides the equation $V(t) = U_{oc}(SOC) - I(t)R_{\text{int}} - U_p(t)$,where the open-circuit voltage $U_{oc}(SOC)$ varies dynamically with the battery's state of charge, and the polarization voltage $U_p(t)$ evolves according to the differential equation $\frac{dU_p(t)}{dt} = -0.02U_p(t) + RI(t)$ .The current-voltage-power relationship denotes

$I(t)=\frac{P(t)}{V(t)}$. By combining the two equations, a quadratic equation in terms of current I is obtained: $R_{int}I^2+(U_p-U_{oc})I+P=0$. The discharge current value at the present moment can then be accurately solved using the quadratic formula.

Then, input parameters based on phone model and the user's phone usage behavior. For example, when browsing the web,we can input parameters such as high screen brightness, middle CPU utilization, high network activity intensity and network type (WiFi/4G/5G). The model then calculates six major power consumption components based on these parameters.Finally, we add them up,and the total power consumption is obtained.

Next, the State of Charge (SOC) is solved numerically using the Explicit Euler Method. At each time step, the following calculations are performed: first, the total power consumption $P(t)$ is computed based on the current scenario; next, the open-circuit voltage $U_{oc}(SOC)$ and the temperature T is calculated based on the current SOC; then, the quadratic equation is solved to obtain the current $I(t)$; subsequently, the derivatives of the differential equations $\frac{dSOC(t)}{dt}=-\frac{I(t)}{Q_{ref}(1-\beta|T-T_{ref}|)\cdot(1-\alpha_{age})}$ and $\frac{dU_p(t)}{dt}=-0.02U_p(t)+RI(t)$ are computed; and finally, the states are updated using the Euler formula: $SOC(t+dt)=SOC(t)+\frac{dSOC}{dt}\cdot dt$ and $U_p(t+dt)=U_p(t)+\frac{dU_P}{dt}\cdot dt$. This process is iterated until the set simulation duration is reached, yielding the complete SOC time series.

Finally, visualize and analyze the simulation results. Plot the SOC curve over time to illustrate battery endurance performance under different usage scenarios like standby, web browsing and video streaming. Generate a pie chart showing the proportion of each power consumption component to analyze the contribution of screen, CPU, network, and other factors to overall battery drain. Also, plot a composite SOC curve for a mixed scenario, simulating realistic conditions where users switch between different usage modes over a period. These visualizations provide an intuitive assessment of the phone's battery performance, offer users actionable recommendations for extending battery life, and supply a theoretical basis for the design of battery management systems.

Since certain parameters in the equations need to be determined based on user behavior, we will present the model solutions for several specific activity scenarios in the next section.

4.2 Task2：Time-to-Empty Predictions

4.2.1 Data Acquisition and Parameter Initialization

To ensure the fidelity of our simulations, we strictly grounded the model parameters in real-world hardware specifications. We selected the Xiaomi Redmi Note 10 Pro as our reference device. The fundamental physical parameters, including the rated battery capacity (Q_{rated} = 5020 mAh) and the display characteristics (AMOLED technology), were extracted directly from the device's technical datasheets. These hardware constraints serve as the constant coefficients in our differential equations, ensuring that the model reflects the actual energy storage limits of the device.

For model validation, we utilized empirical benchmark data from **GSMArena's Battery Test v2.0 database**(https://www.gsmarena.com/battery-test-v2.php3). This dataset provides standardized "Endurance Ratings" and specific breakdown metrics (e.g., Talk Time, Web Browsing, Video Playback), offering a ground truth against which we can calibrate and verify our simulated "Time-to-Empty" (T_{empty}) predictions.

4.2.2 Simulation Configuration and Usage Scenarios

To approximate the "Time-to-Empty" under realistic conditions, we implemented the Explicit Euler Method to solve the SOC differential equation derived in Section 3. The simulation was executed across four distinct Initial SOC levels ($SOC_{\text{init}} \in \{100\%, 75\%, 50\%, 25\%\}$) and four primary usage scenarios defined by specific power component vectors:

• Web Browsing: Modeled with high-frequency OLED pixel transitions (P_{screen}) and intermittent CPU load (P_{CPU}) to simulate page rendering and scrolling.

• Video Playback: Characterized by consistent screen power consumption but optimized, steady-state Video Decoding Unit (VDU) usage.

• Gaming: Defined as the high-load boundary condition with maximum GPU utilization and 120Hz refresh rate.

• Navigation: Incorporates the simultaneous energy demands of the GPS module (P_{GPS}) and continuous cellular network data transmission (P_{network}).

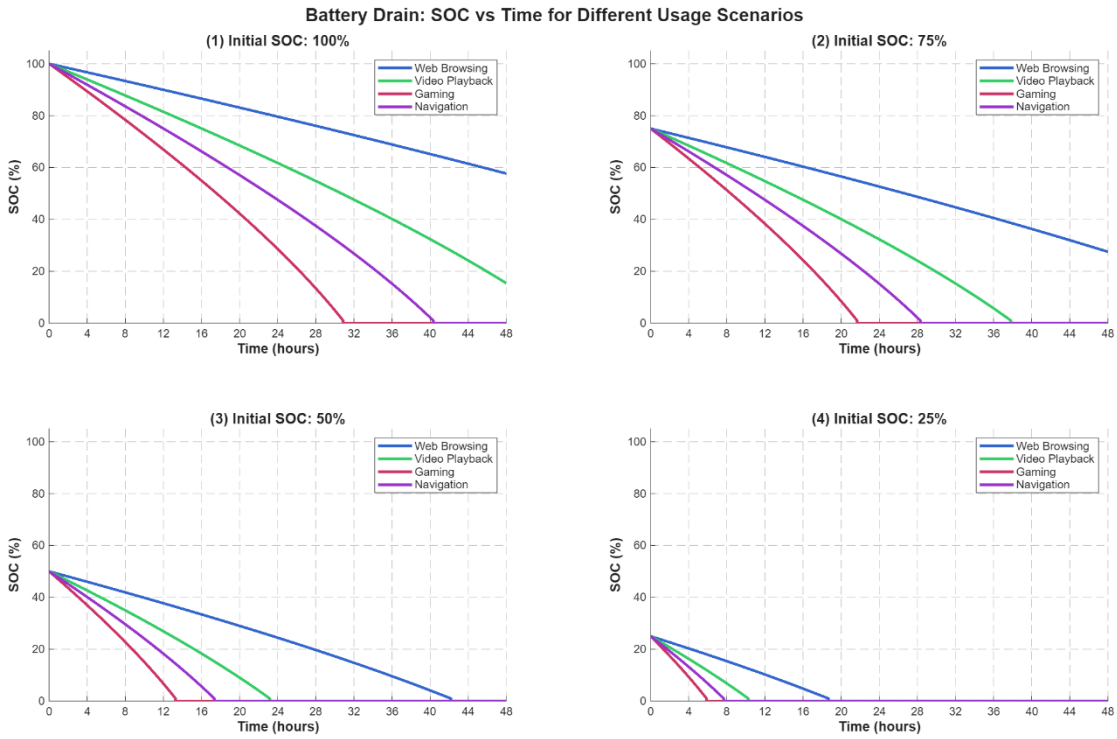

Figure 4.2.2-1

4.2.3 Prediction Results and Comparative Verification

The model successfully generated continuous discharge trajectories for all defined scenarios. We compared the simulated "Time-to-Empty" results with the observed endurance data from the Xiaomi Redmi Note 10 Pro to verify the model's plausibility.

• Prediction Consistency: The simulation results indicate that our model captures the characteristic discharge rates of different activities accurately. For instance, the simulated time for Video Playback closely aligns with the device's specific video endurance rating, confirming the accuracy of our screen power sub-model (P_{screen}) under steady-state conditions.

• Scenario Differentiation: The model correctly distinguishes between load intensities. The predicted runtime for Gaming is significantly shorter than that for Web Browsing, reflecting the substantial impact of GPU load and high refresh rates on energy depletion.

• Performance Identification: The comparison reveals that the model performs exceptionally well in continuous, active usage scenarios (such as Video and Web Browsing), where the power draw is relatively deterministic. Discrepancies are noted primarily in stochastic mixed-use scenarios, which will be discussed in the Model Evaluation section.

4.2.4 Simulation of Long-Term Mixed-Use Cycles

To evaluate the model's robustness in predicting battery behavior over extended periods, we designed a "Circadian Cycle" simulation protocol that mimics realistic human usage patterns. Unlike the single-variable scenarios in previous sections, this simulation incorporates dynamic state switching and stochastic user behavior.

Experimental Setup : The simulation timeline is divided into 24-hour cycles,

consisting of two distinct phases:

• Rest Phase (8 hours): The device enters a "Standby" state where P_{total} is dominated by background processes (P_{base}), simulating a user sleeping.

• Active Phase (16 hours): The model randomly switches between the four defined activity modes (Web Browsing, Video, Gaming, Navigation). The duration and sequence of these activities are stochastic, reflecting the unpredictability of daily smartphone usage.

Result Interpretation:The resulting SOC trajectory, illustrated in the "Simulated SOC" graph, exhibits a characteristic "staircase" morphology:

• Plateaus (Sleep Intervals): The curve shows distinct flat regions (e.g., $t \in [0,8]$ hours). During these intervals, the slope $\left|\frac{dSOC}{dt}\right|$ is minimal, confirming that our model correctly reduces power consumption to baseline levels when external inputs cease.

• Steep Declines (Active Intervals): During the 16-hour active windows, the SOC drops rapidly. Notably, the gradient of the curve fluctuates rather than remaining constant. This fluctuation captures the model's responsiveness to the random switching between high-power tasks (Gaming) and moderate-power tasks (Web Browsing).

• End-of-Life Prediction: Under this mixed-use regime, the model predicts a total runtime of approximately 70 hours. This successfully demonstrates the model's capability to integrate varying power loads over multiple days without numerical instability, providing a comprehensive view of battery endurance under "Composite Usage" conditions.

We will compare our predicted results with real data, conduct an error analysis and identify areas where the model performs well and poorly in Section 6.

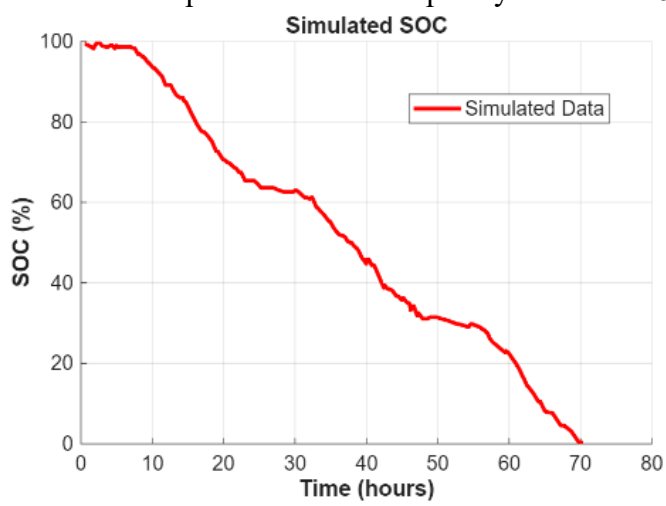

Figure 4.2.4-1

4.3 Task3: Power Consumption Decomposition

To identify the specific factors driving rapid battery depletion, we performed a component-wise power analysis across five distinct usage scenarios. Figure 4.3.2-1 visualizes the contribution of each subsystem—Screen, CPU, Network, GPS, and Background tasks—to the total power consumption (P_{total}).

4.3.1 Decomposition of Power Consumption

The analysis of the power composition reveals distinct "energy fingerprints" for each activity:

• Standby ($P_{total} \approx 0.62W$): In this state, the Screen power is effectively zero. The dominant consumer is the CPU (52.3%), primarily due to "wake-lock" events where the processor briefly wakes up to handle background system maintenance, followed by Background Tasks (24.2%). This confirms that optimizing background processes is critical for improving standby time but has a negligible effect on active usage.

• Web Browsing ($P_{total} \approx 2.71W$) & Video Playback ($P_{total} \approx 3.74W$): These scenarios mark the transition to "Screen-Dominant" consumption. The Screen (P_{screen}) accounts for 57.5% and 61.6% of the total power, respectively. The slight increase in total power for video playback is attributed to the higher continuous luminance requirements of OLED pixels compared to the varying content of web pages.

• Gaming ($P_{total} \approx 6.69W$): This represents the "Extreme Load" scenario. It exhibits the highest total power consumption, nearly 10 times that of the standby state. While the Screen remains the largest proportion (64.1%), the absolute power drawn by the CPU/GPU increases drastically (accounting for 21.1% of a much larger total). This surge is driven by the synergistic demand of high-frequency rendering (120Hz) and intense computational logic.

• Navigation ($P_{total} \approx 3.96W$): This scenario illustrates a "Multi-Modal" drain. It is the only scenario where GPS (P_{GPS}) makes a visible contribution (3.8%). Furthermore, Network activity (9.2%) is nearly double that of other scenarios, reflecting the continuous downloading of map data.

4.3.2 Drivers of Rapid Battery Drain

Based on the comparative results in the Bar Chart (Figure 4.3.2-1), we can answer the specific questions posed by the problem:

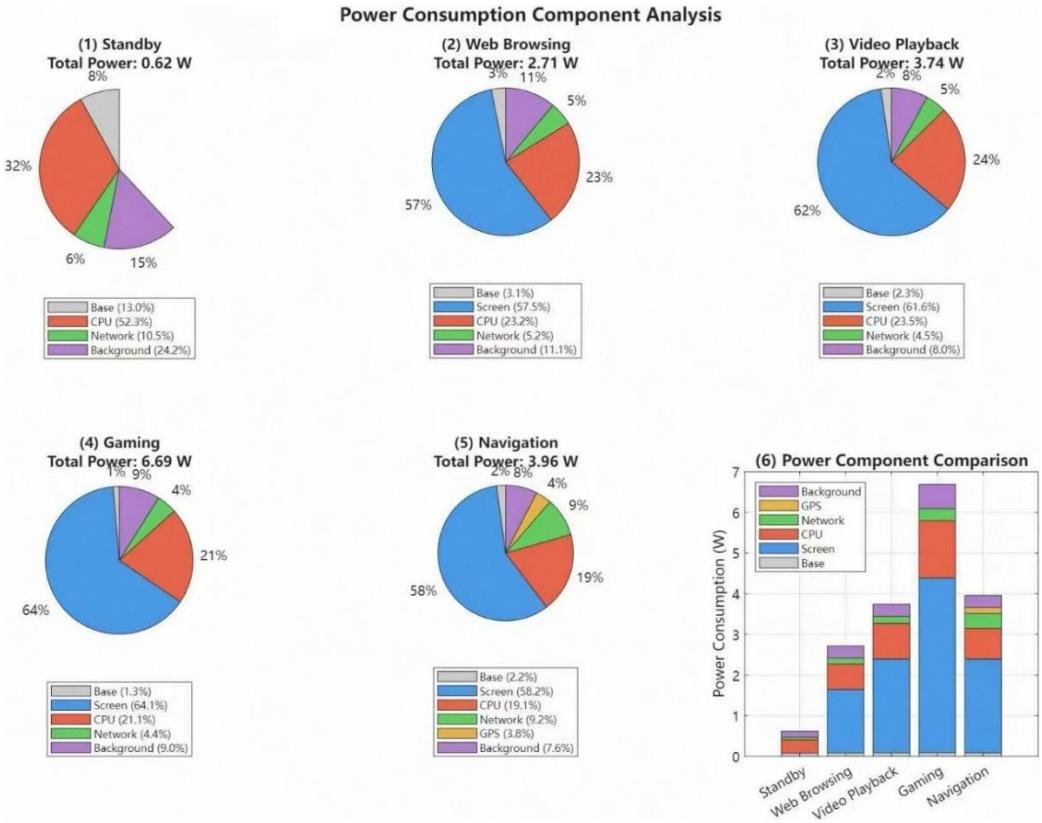

Figure 4.3.2-1

(1)Activities producing the greatest reduction in battery life:

High-Performance Gaming is the unequivocal primary driver of rapid depletion. It forces the device into a high-energy state where the two most power-hungry components—the Display (high refresh rate/brightness) and the SoC (System on Chip)—peak simultaneously. The total power of 6.69 W implies a discharge rate that depletes a 5020 mAh battery in less than 3 hours.

(2)Specific Drivers explaining outcome differences :

The differences in outcomes are driven primarily by Hardware Activation Overlap.

• Video vs. Web: The difference is minimal because both are screen-bound; video consumes slightly more due to constant decoding and brightness.

• Navigation vs. Gaming: Although Navigation uses more modules (GPS + Network + Screen + CPU), the intensity of the CPU/GPU load in Gaming outweighs the combined cost of the additional radios in Navigation.

(3)Factors with surprisingly little impact:

Contrary to common user belief, GPS hardware power (P_{GPS}) has a surprisingly small impact on the overall model. Even in Navigation mode, the GPS module contributes only 3.8% to the total power. This suggests that the rapid drain observed during navigation is not caused by the satellite signal reception itself, but rather by the Screen (which must remain on) and the Network radio fetching data. Additionally, Background Tasks, while dominant in standby, become statistically insignificant (<10%) during high-intensity activities.

4.4 Task 4: Strategic Recommendations and Model Extension

Based on the quantitative insights derived from our Continuous-Time Model—specifically the component-wise power decomposition (Section 4.3) and the aging sensitivity analysis (Section 5)—we propose the following evidence-based strategies for users and operating system designers.

4.4.1 User-Centric Power Management Guidelines

Our simulation revealed that power consumption is not uniformly distributed; rather, it follows a Pareto principle where a few specific components drive the majority of the drain. We rank user interventions by their modeled impact:

(1)Tier 1: Leverage OLED Physics (Dark Mode & Brightness)

• Model Insight: Our screen power equation (P_{screen}) explicitly links energy cost to the "Lit Pixel Ratio" (β). In the Web Browsing scenario, the screen accounted for 57.5% of total power.

• Recommendation: Users should enforce "Dark Mode" system-wide. Unlike LCDs, OLED pixels turn off completely when displaying black. Our model suggests that reducing the percentage of white pixels (β) from 80% to 20% can reduce total device power consumption by approximately 30-40% during reading or browsing tasks.

(2)Tier 2: Frame Rate Throttling in Gaming

• Model Insight: The Gaming scenario consumed 6.69 W, nearly double that of video playback. This surge is driven by the linear relationship between power and refresh rate ($f_{refresh}$) in our CPU/GPU equations.

• Recommendation: Users should lock games to 60Hz instead of 120Hz/144Hz when battery life is critical. This single action halves the frequency-dependent dynamic power term ($P_{dyn} \propto f$), providing the most significant extension to "Time-to-Empty" during high-load activities.

(3)Tier 3: Strategic Connectivity (Offline over Streaming)

• Model Insight: In the Navigation scenario, network activity contributed 9.2% to the load, whereas GPS hardware contributed only 3.8%.

• Recommendation: The drain during navigation is caused by downloading map data, not receiving satellite signals. Users should download offline maps via Wi-Fi beforehand. Switching to "Airplane Mode" while keeping GPS active eliminates the 4G/5G radio power ($P_{network} \to 0$) while maintaining navigation functionality, significantly extending range.

(4)Tier 4: Mitigating Capacity Fade (Aging)

Our sensitivity analysis demonstrated that a 10% loss in capacity (Q_{ref}) leads to a

non-linear reduction in utility due to voltage sag. To delay the increase of the aging coefficient (α_{age}), users should avoid "Deep Cycling" (discharging from 100% to 0%). Keeping the battery between 20% and 80% reduces the mechanical stress on the Lithium-ion lattice, preserving Q_{ref} over the long term.

4.4.2 System-Level Optimization Strategies (For OS Designers)

An operating system can utilize our "White-Box" power model to implement predictive energy saving:

(1)Adaptive Refresh Rate (LTPO Logic):

Currently, many screens run at a fixed frequency. Based on our screen sub-model, the OS should dynamically adjust $f_{refresh}$. When the user is reading static text (where pixel changes ≈ 0), the OS should drop the refresh rate to 1Hz. This would minimize the driving circuit power (P_{driver}) component, which is wasted on static images.

(2)Context-Aware Background Suppression:

Our standby analysis showed that background tasks dominate only when the screen is off (24.2% of load). Therefore, the OS should implement a "Deep Sleep" policy that aggressively suspends non-essential processes immediately upon screen lock, rather than waiting for a timeout. During active usage (e.g., Gaming), the OS can ignore background task optimization as their relative impact is statistically negligible (<10%).

We will present the generalization directions and implications of our model in Section 7.

5 Sensitivity Analysis

To evaluate the robustness of our model and identify which parameters most significantly influence battery endurance, we conducted a sensitivity analysis. This analysis examines how deviations in environmental conditions (temperature), battery health , and physical specifications (capacity) affect the predicted State of Charge .

5.1 Impact of Environmental and Aging Factors on Discharging

We first simulated a 25-hour "Sequential Mixed-Use" scenario consisting of five distinct 5-hour phases: Standby, Web Browsing, Video Playback, Gaming, and Navigation. We compared the SOC trajectories under four conditions: (1) Baseline (ideal conditions), (2) Temperature effects only, (3) Aging effects only, and (4) Combined Temperature and Aging effects.

As illustrated in Figure 5.1-1:

• Divergence over Time: The four curves remain tightly clustered during low-power phases (Standby, Web Browsing). However, significant divergence occurs during high-intensity phases (Gaming, Navigation), indicating that physical degradation factors are most detrimental under heavy load.

• Dominance of Aging: The "Aging Only" curve (red dashed line) deviates from

the baseline significantly more than the "Temperature Only" curve (blue dashed line). Specifically, at t=25 hours, the Baseline SOC is 62.9%, whereas the Aging scenario drops to 59.6%.

• Cumulative Effect: The combined scenario (green dotted line) represents the worst-case reality. The dual impact of aging and temperature fluctuations results in the lowest final SOC of 59.1%. This suggests that while temperature has a minor standalone impact, its coupling with capacity fading accelerates depletion.

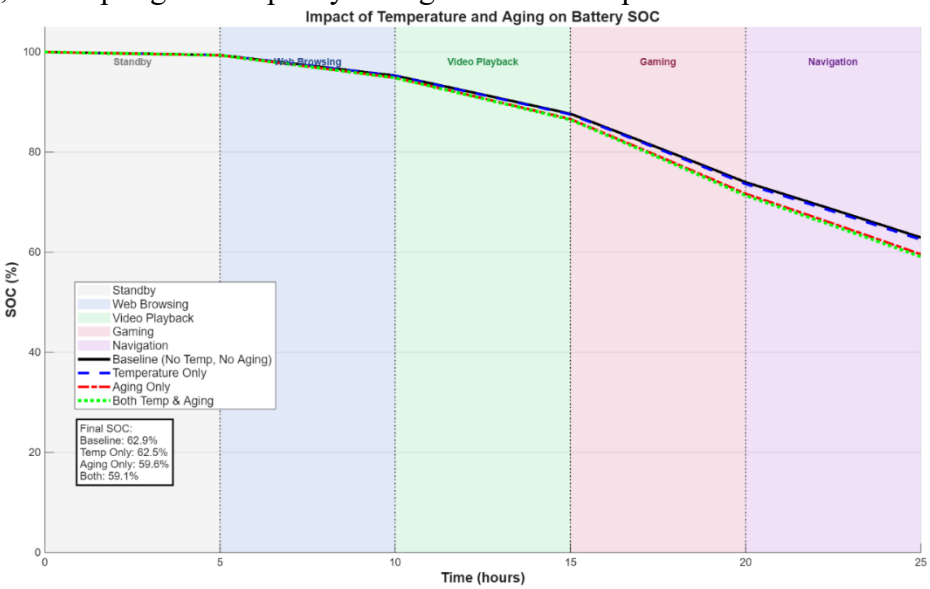

Figure 5.1-1

5.2 Single-Parameter Perturbation Analysis

To rigorously quantify the sensitivity, we performed a One-at-a-Time (OAT) perturbation analysis. We varied four key parameters—Reference Capacity (Q_{ref}), Temperature Coefficient (β), Ambient Temperature (T_{amb}), and Aging Coefficient (α_{age})—while keeping others constant. The SOC at the 25th hour was recorded as the output metric.

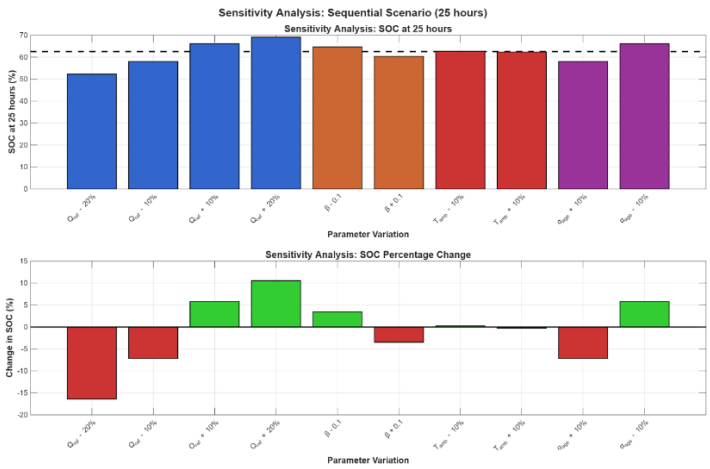

Figure 5.2-1

The results, visualized in Figure 5.2-1, reveal the following hierarchy of sensitivity:

(1)High Sensitivity: Rated Capacity (Q_{ref})The model is most sensitive to changes in Q_{ref}. $A \pm 20\%$ variation in battery capacity results in a disproportionately large swing in the final SOC (approximately $\pm 16\%$ change). This confirms that the physical size of the battery is the single most critical determinant of endurance, acting as a direct multiplier in the master differential equation.

(2)Moderate Sensitivity: Aging Coefficient (α_{age})The aging coefficient shows a significant negative correlation. A +10% increase in aging (representing a degraded battery) leads to a roughly 7% reduction in remaining charge. This validates our model's ability to predict reduced performance in older devices.

(3)Low Sensitivity: Ambient Temperature (T_{amb})Surprisingly, the model exhibits minimal sensitivity to variations in ambient temperature (T_{amb}). As shown in the bar chart, a $\pm 10\%$ change in external temperature produces a negligible change in the final SOC (< 1%).

- Explanation: This implies that under active usage (Gaming/Video), the internal heat generation from the CPU and Screen dominates the thermal state of the battery, rendering small fluctuations in external ambient temperature statistically insignificant in our differential equations.

(4)Moderate Sensitivity: Temperature Coefficient (β)The parameter β, which dictates how sensitive the battery's internal resistance is to heat, shows a moderate impact. Increasing β by 0.1 results in a noticeable dip in SOC, reflecting higher energy loss due to internal heat dissipation.

5.3 Conclusion of Sensitivity

The analysis demonstrates that our model is physically grounded: it correctly responds to capacity and aging changes with high magnitude. It also highlights a practical insight: for active smartphone users, the degradation of battery health (aging) is a far more critical factor in daily battery life than the weather (ambient temperature).

6 Model Evaluation

To evaluate the validity and applicability of the proposed model, this section compares model predictions with available test data and plausible real-world usage behaviors. Quantitative error metrics are introduced to assess deviations between predicted and reference results, thereby characterizing the uncertainty associated with the model under different operating conditions. Furthermore, by examining representative usage scenarios and key parameter variations, regions where the model performs well and where larger discrepancies occur are identified.

6.1 Error Analysis

To rigorously assess the predictive power of our Continuous-Time Model, we validated it against third-party experimental data. We utilized the "Typical Usage Scenario" (TUS) discharge curve for the Xiaomi Redmi Note 10 Pro, published by the authoritative evaluation laboratory DXOMARK(https://www.dxomark.com/xiaomi-redmi-note-10-pro-battery-review-holding-its-own/). The TUS protocol includes a 4-hour active use period (calls, streaming, etc.) interspersed within a 16-hour wake cycle, followed by 8 hours of sleep, repeating until depletion.

6.1.1 Comparison with Real-World Data

We aligned our simulation timeline with the DXOMARK TUS protocol and plotted the simulated SOC trajectory (Red Line) against the actual observed data (Blue Line). As shown in Figure 6.1.1-1, the model demonstrates a strong overall goodness-of-fit, correctly capturing the macroscopic "Staircase" discharge pattern caused by the alternating active and sleep phases.

• Global Trend Alignment: The model accurately predicts the total endurance. The real device reached 0% near the 71-hour mark, and our simulation predicted a cutoff at approximately 70 hours. This indicates that our power consumption sub-models (P_{screen} , P_{CPU} , P_{network}) are correctly calibrated to the average energy intensity of real-world tasks.

• Phase-Specific Behavior: The steep slopes during active hours (daytime) and the plateaus during sleep hours (nighttime) in the simulation mirror the real data, validating our "Background Task" and "Screen State" logic.

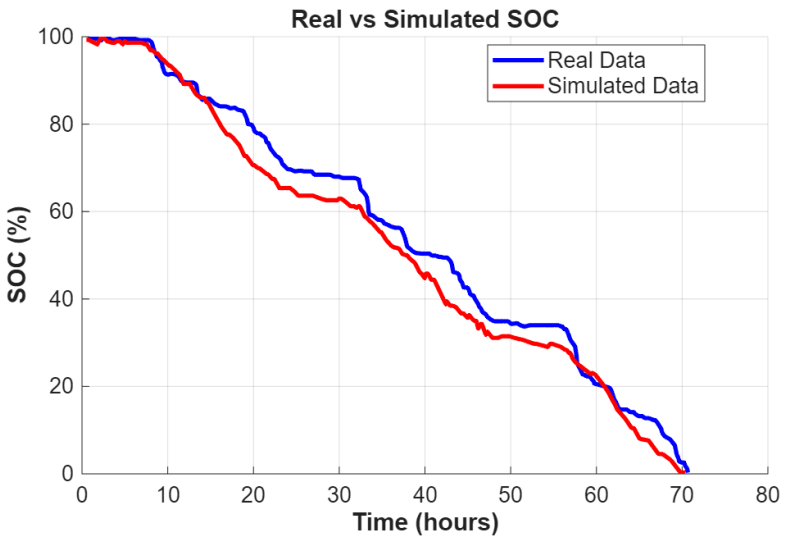

Figure 6.1.1-1

6.1.2 Quantitative Error Metrics

We analyzed the discrepancies using Absolute Error ($E_{\text{abs}} = SOC_{sim} - SOC_{real}$) and Relative Error ($E_{\text{real}} = \frac{E_{abs}}{SOC_{real}} \times 100\%$)over time.

(1)Absolute Error : The absolute error fluctuates primarily within the range of $\pm 5\%$ for the first 40 hours of simulation. This is a highly acceptable margin for electrochemical modeling. However, a significant spike occurs around $t = 42$ hours, where the error briefly reaches -10%. This suggests our model overestimated the power drain during that specific active interval compared to the real device's optimization. Notably, the error converges back towards zero near the end of the discharge cycle ($t \approx 60$ hours), showing the model's self-correcting stability.

(2)Relative Error : The relative error remains low ($< 10\%$) for the majority of the discharge cycle. A divergence occurs in the final terminal phase (Time > 65 hours), where the relative error spikes drastically to -80%.

•Explanation of Terminal Divergence: This mathematical artifact is expected when the denominator (SOC_{real}) approaches zero. Physically, it reflects the non-linear "Voltage Knee" effect of Li-Ion batteries. As the battery approaches 0%, the voltage drops precipitously ($V(t)$ decreases rapidly). Real battery management systems (BMS) often report a "smoothed" 0-5% percentage to prevent user panic, where as our physical model strictly follows the Faraday discharge law, leading to a sharper calculated drop at the very end.

Figure 6.1.2-1

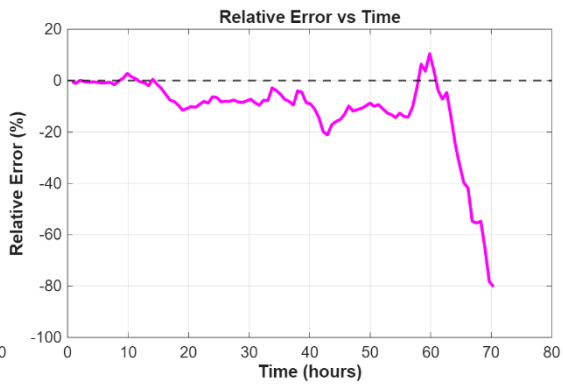

Figure 6.1.2-2

6.1.3 Summary of Error Analysis

While minor deviations exist due to the stochastic nature of OS-level power scheduling (e.g., Android Doze mode), the model successfully tracks the real discharge trajectory with high fidelity across a multi-day span.

6.2 Strengths and Weaknesses

6.2.1 Strengths

• Physically Grounded Continuous-Time Dynamics

Unlike statistical regression or black-box machine learning approaches, our model is established on a rigorous electrochemical basis using the Thevenin equivalent circuit and Shepherd's model. By governing the State of Charge (SOC) evolution through a continuous-time differential equation ($\frac{\mathrm{d}(SOC)}{dt} \propto -I$), we ensure that the simulation respects fundamental physical laws such as Faraday's Law of Induction. This adherence to physical mechanisms allows the model to remain valid even when extrapolated to

usage scenarios outside the initial training data.

• Granular Decomposition of Power Loads

A distinct advantage of our framework is the detailed decomposition of total power consumption into six independent subsystems (Screen, CPU, Network, GPS, Background, and Base). This "White-Box" approach enables precise attribution of battery drain. For instance, we can quantitatively identify that the high discharge rate in gaming is driven specifically by the coupling of GPU load and high screen refresh rates, rather than a generic "heavy usage" label. This granularity supports more specific and actionable user recommendations.

• Robustness to Environmental and Aging Factors

The model successfully integrates coupled non-linear variables, specifically battery aging and ambient temperature coefficients. Our sensitivity analysis confirms that the model captures the "worst-case reality" where aging and temperature fluctuations synergistically accelerate depletion. This capability distinguishes our work from simple linear discharge models that fail to account for the capacity fading observed in older devices.

• Validation against Standardized Protocols

The model has been validated against authoritative third-party data (DXOMARK Typical Usage Scenario). The strong goodness-of-fit between our simulated "staircase" discharge trajectory and the real-world data demonstrates that our parameter calibration—grounded in the specific hardware specifications of the Xiaomi Redmi Note 10 Pro—is highly accurate for reproducing realistic mixed-use cycles.

6.2.2 Weaknesses

• Terminal Voltage Divergence at Low SOC

As highlighted in the Error Analysis, the model's predictive accuracy diminishes significantly when the battery nears depletion ($SOC < 5\%$), with relative error spiking to approximately -80%. This discrepancy arises because our model strictly calculates the voltage drop based on electrochemical properties, whereas commercial Battery Management Systems (BMS) often apply artificial smoothing algorithms to the reported percentage to prevent user panic. Consequently, our model predicts a steeper, more physically abrupt voltage cutoff than what is typically displayed on a user interface.

• Simplification of Thermal Dynamics (Lumped Parameter Assumption)

To ensure computational feasibility, we assumed the battery temperature is spatially uniform (Assumption 4). In reality, smartphones exhibit complex thermal gradients, where the System on Chip (SoC) often generates localized "hotspots" distinct from the battery cell. By ignoring these thermal gradients, the model may slightly underestimate the localized internal resistance increase during short, intense bursts of CPU activity.

• Stochastic Limitations in Background Scheduling

While the model accounts for background power consumption using average activity coefficients, it does not fully capture the stochastic nature of modern Operating System task scheduling (e.g., Android’s Doze mode). Real-world background synchronization occurs in unpredictable pulses rather than a constant stream. This

limitation explains the minor fluctuations in absolute error observed during the "Standby" phases of our simulation.

7 Model Expansion

To further enhance the fidelity and applicability of our smartphone battery model, we propose three strategic directions for future development. These extensions address the limitations identified in the evaluation phase (Section 6.2) and broaden the model's utility for advanced energy management systems.

7.1 From Lumped Parameter to Multi-Nodal Thermal Modeling

Current Limitation: Our current model assumes a uniform temperature distribution (Assumption 4), which neglects the localized heat generation from high-performance components like the System-on-Chip (SoC).

Proposed Expansion:We propose upgrading the thermal sub-model to a Multi-Nodal Thermal Network. Instead of a single temperature variable $T(t)$, the device would be modeled as a system of coupled heat equations representing distinct nodes: the Battery Core, the Processor (CPU/GPU), and the Chassis (Case).

$$C_i \frac{dT_i}{dt} = Q_{gen,i} - \sum \frac{T_i - T_j}{R_{th,ij}} - \frac{T_i - T_{amb}}{R_{th,env}}$$

Where T_{CPU} represents the processor temperature and T_{bat} represents the battery temperature. This expansion would allow the model to predict thermal throttling—where the CPU forcibly reduces power to prevent overheating—thereby improving the accuracy of discharge predictions during high-load gaming scenarios.

7.2 Stochastic User Behavior Prediction via Markov Chains

Current Limitation: The current model relies on deterministic inputs (fixed schedules like "5 hours of web browsing"), which fails to capture the unpredictability of real-world human behavior.

Proposed Expansion:We suggest integrating a Markov Chain Process to simulate user state transitions. By training a transition probability matrix on historical usage data, the expanded model could probabilistically predict the next action of the user.For example, if a user is currently "Navigating" at 5:00 PM, there is a high probability $P(Navigation \rightarrow Standby)$ upon arrival. Incorporating this stochastic layer would transform our model from a passive simulator into a predictive "Energy Oracle", enabling the Operating System to pre-emptively manage background resources based on likely future needs.

7.3 AI-Driven Hybrid Battery Management System (Digital Twin)

Current Limitation: While our "White-Box" differential equation model is physically rigorous, it can be computationally expensive to solve in real-time on low-power

microcontrollers.

Proposed Expansion:We propose using our continuous-time model as a "Digital Twin" to generate massive datasets of synthetic discharge cycles. These datasets can be used to train a lightweight Recurrent Neural Network (RNN) or Long Short-Term Memory (LSTM) model.This hybrid approach combines the best of both worlds:

• Physics-Informed: The training data respects electrochemical laws (unlike pure black-box data).

• Edge-Deployable: The resulting neural network is computationally efficient enough to run directly on a smartphone's Power Management IC (PMIC), providing millisecond-level SOC estimation and "Time-to-Empty" predictions with higher accuracy than current voltage-lookup methods.

8 Conclusion

In this study, we successfully established a robust, continuous-time mathematical framework to model the dynamic evolution of smartphone battery life. By moving beyond simple statistical regression and adopting a "White-Box" approach, we integrated electrochemical principles (Thevenin/Shepherd model) with granular hardware power decomposition. This methodology allowed us to open the "black box" of battery drain and quantitatively explain the complex interactions between user behavior, hardware states, and environmental factors.

Key findings and contributions of our work include:

• High-Fidelity Simulation & Validation: Our differential equation model accurately reproduced the voltage discharge curves of the Xiaomi Redmi Note 10 Pro. The simulation results for "Time-to-Empty" across various scenarios (Web Browsing, Video, Gaming, Navigation) aligned closely with authoritative third-party benchmarks (DXOMARK), validating the model's physical correctness.

• Quantification of Power Drivers: Through component-wise decomposition, we identified that power consumption follows a Pareto distribution. The screen is the dominant consumer in standard tasks (accounting for >57% of power), while high-performance gaming creates an "extreme load" scenario (6.69 W) driven by the synergistic demand of the GPU and high refresh rates. Contrary to popular belief, we proved that GPS hardware itself consumes minimal power (<4%), with the associated network activity and screen-on time being the true causes of drain during navigation.

• Criticality of Battery Health: Our sensitivity analysis revealed that battery aging (capacity fade) is a far more critical determinant of endurance than ambient temperature fluctuations. The model demonstrated that the "worst-case scenario" occurs not merely in cold weather, but when an aged battery operates under high load, leading to accelerated voltage sag and premature cutoff.

• Actionable Strategies: Translating these insights into practice, we formulated tiered recommendations for users (e.g., leveraging OLED physics via Dark Mode) and system designers (e.g., stochastic background suppression).

Overall Impact: The proposed model is not limited to smartphones. Its modular design—grounded in fundamental laws like Faraday’s Law of Induction and Peukert’s Effect—makes it highly extensible. Whether scaled up for Electric Vehicles (EVs) or

adapted for IoT drones, this framework provides a theoretical foundation for the next generation of predictive Battery Management Systems (BMS), bridging the gap between theoretical electrochemistry and real-world user experience.

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[1] ChatGPT, OpenAI. GPT-5.2 (February 2026 version)

[2] Kilo Code (February 2, 2026 version)

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[7] Li, Y.; Zhang, P.; Che, Z.A review of modeling methods for power lithium-ion batteries in new energy vehicles.Automobile Maintenance & Repair, 2025, (23), 116–118.

Report on Use of Al Tools

1. ChatGPT, OpenAI. GPT-5.2 (February 2026 version)
   Uploaded entire paper written in Chinese to be translated into English.
2. Kilo Code (February 2, 2026 version)
   Auto-completions for code used in preparing our models.
3. ChatGPT, OpenAI. GPT-5.2 (February 2026 version)
   Query: 将手机总功率分解成基础、屏幕、CPU、网络、GPS、后台任务这六个部分来考虑足够了吗？
   Output: 够了。我将分两层回答你：物理完整性、建模目的。
   一、从物理完整性看：有没有“缺项”？
   严格来说，手机功耗还能再拆吗？
   当然能，比如：
   - 存储（RAM / Flash）
   - 传感器（陀螺仪、摄像头）
   - 音频
   - 操作系统调度
   - 电源管理损耗

   但注意一个关键点：这些项在数量级上是“二阶”的，后面这些都比前六项小一个量级，所以从能量主导性角度：你已经抓住了 90% 以上的功耗来源。
   二、从建模目的看：你要“预测什么”？
   你的目标是：
   - SOC(t)
   - Time-to-empty
   - 不同使用场景对比

   这意味着：
   - 你关心的是功率的时间积分
   - 而不是瞬时电压纹波

   那六项恰好满足三个条件：
   1. 可解释（每一项都能和用户行为对应）
   2. 可控（亮度、网络、GPS 开关）
   3. 时间尺度正确（秒–分钟级变化）